

Combinatorics Practice Problems- Solutions

1. A family of eight people want to go to the airport. They call two taxi cabs. In how many different ways can they get on the cabs if:

Each taxicab has room for a maximum of five passengers.

Which chair a person sits on or next to whom they sit does not make a difference.

Which cab a person rides on does not make a difference.

Which group of people a person rides with is important and we would like to count the different possible combinations of that.

Note that we are looking for a solutions that satisfies all the above requirements.

The possibilities for getting on the cabs are:

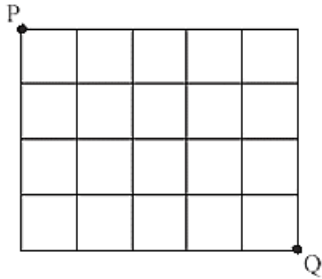
- 1) A group of 3 and a group of 5
- 2) A group of 4 and a group of 4

$$C(8,3) + \frac{C(8,4)}{2} = 56 + 35 = 91$$

Notes:

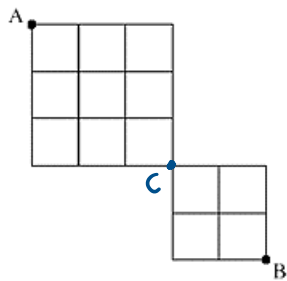
- 1) we are not counting "A group of 5 and a group of 3" separately since which cab they take does not make a difference and it is already counted in "A group of 3 and a group of 5".
- 2) We need to divide $C(8,4)$ by 2 because when we count all choices of 4 among 8, these two case are counted as separate cases: Case 1: ABCD-(EFGH) Case 2: EFGH- (ABCD) whereas they should be counted as one case since they are the same group of people taking a ride to the airport.

2. Moving only right or down, how many paths are there from P to Q in the first grid? How many paths from A to B in the second grid?



Each path from P to Q would be a permutation of 5R's (R= right) and 4 D's (D = Down). Use MISSISSIPPI formula:

$$\frac{9!}{4! 5!} = 126$$



Before the bottleneck, 3R's and 3D's: $\frac{6!}{3!3!} = 20$

After the bottleneck: 2R's and 2D's: $\frac{4!}{2!2!} = 6$

To get from A to B: $20 \times 6 = 120$

3. In what percent of 12-digit strings that can be made with 4A's, 4B's, and 4C's all of the A's are next to each other?

All: $\frac{12!}{4!4!4!} = 34650$ (MISSISSIPPI problem)

The ones with A's together:

Tie up the A's together. They become one object. Along with 4B's and 4C's, we have 9 objects:

$$\frac{9!}{4!4!} = 630$$

The percentage of the sequences asked for: $\frac{630}{34650} = \frac{1}{55} = 1.8\%$

4. How many 5-digit integers have no 9's ? Note that 00087 is not a 5-digit integer (it is a 2-digit integer), but 80770 is a 5-digit integer.

All 5-digit integers without 9: $8 \times 9^4 = \mathbf{52488}$

Note that for the highest place value, we have 1, 2, 3, 4, 5, 6, 7, 8 (no zeros, no 9's)

For the remaining place values, we have 0, 1, 2, 3, 4, 5, 6, 7, 8 (no 9's)

5. In a class of 30 students, how many committees of 3 people can be selected if the roles they are selected for is the same? $C(30, 3) = 4060$

What if the roles are different?

$$P(30, 3) = 24360$$

6. When you play lotto 6-49, you choose 6 different integers from 1 to 49 inclusive. The order in which the numbers are chosen is not important. How many combinations are possible?

If the order at which the number were picked would matter, how many choices would be possible?

The order is not important:

$$C(49,6) = 13983816$$

The order is important:

$$P(49,6) = 10068347520$$

7. Assume a car license plate consists of 7 characters. The first 3 can be A to F (inclusive, uppercase) but no letter can be repeated. The next 3 characters can be any of the digits from 1 to 9 (inclusive) with possible repetition. The last character can be any of the letters X, Y, Z.

How many license plates are possible?

Example: ABF 119 X

$$P(6,3) \times 9^3 \times C(3,1) = 120 \times 729 \times 3 = 262440$$

8. What percent of different arrangements of all the letters of the word OOMPALOOMPA have all the O's together? Example: MPALOOOOMPA

O O O O

M M

P P

A A

L

All arrangements: $\frac{11!}{4!2!2!2!} = 207900$

Keeping O's together: tie them up, treat them as a new symbol (for example, θ)

θ

M M

P P

A A

L

Number of arrangements: $\frac{8!}{2!2!2!} = 5040$

The percentage of arrangements with all O's together: $\frac{5040}{207900} \times 100\% = \mathbf{2.42\%}$

9. An IPv4- class B address, used for large network consists of 32 binary bits. It starts with 10 followed by a 14-bit network number, followed by a 16-bit local address. Local addresses of all 0's or all 1's are not allowed. There are no restrictions on the network number. How many different IPv4-class B addresses exist?

10

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Starts with 10: 1 possibility, 14-bit network number: $2^{14} = 16384$, 16-bit local address: $2^{16} - 2 = 65534$ (minus 2 because all zeros and all ones are not accepted)

All: $1 \times 16384 \times 65534 = \mathbf{1073709056}$

10. How many hexadecimal strings of length 12: hexadecimal string: we are allowed to use 0-9 and A-F
 a) Have no 9's and have at least one F?

All strings that have no 9's: 15^{12}

All strings without 9 without F: 14^{12}

All strings without 9 who have at least one F: $15^{12} - 14^{12} = 7.305 \times 10^{13}$

- b) Start with 1A1A and contain 9EF (in this order with nothing between 9 and E and nothing between E and F) somewhere in the string? Example: **1A1A0079EF3A**

Starts with 1A1A: one possibility. There remains 8 more bits to be filled. Tie up 9EF and treat as a new symbol (for example θ). We have 6 empty places to fill out with 0-9, A-F, and θ . To keep correct count of bits, we will determine the place of θ first: $C(6,1) = 6$ places for θ to go. Now we fill out the remaining 5 places: $16^5 = 1048576$

So, the answer seems to be $C(6,1) \times 16^5 = 6291456$, however, in this answer, the strings with two 9EF's are double counted, so we will subtract once: $6 \times 16^2 = 1536$

6 because the arrangements with two 9EF's are:

$\theta\theta - -$

$\theta - \theta -$

$\theta - -\theta$

$- \theta - \theta$

$- \theta \theta -$

$- - \theta\theta$

Final answer: $1 \times (6 \times 1048576 - 1536) = 6289920$

11. With A, B, C, M, N, P, 3, 4, 5, 6, 7:

How many passwords of length 7 can we make to meet all of these requirements:

Repetition of letters is allowed, but repetition of digits is not.

The password must start with a letter and must end in a digit.

The password must have exactly 4 letters and exactly 3 digits.

An example: AA76BB3

L - - - - - D

The L and D are fixed, the five middle places are to be 3L's and 2D's.

Finding places for the 3L's and 2D's: $C(5,3) \times C(2,2) = 10$

Now we know where the L's and D's are, consider this arrangement for example: L L D L L D D

Let's fill up these place holders: for L's, we have 6^4 choices and for D's we have $P(5,3)$ choices

Final answer: $10 \times 6^4 \times P(5,3) = 777600$

12. In a film festival ten short movies are to be played in sequence (one after another).

We are tasked with arranging the order in which these short movies are to be played on the screen.

Two of the shorts are in French, the rest are in English.

Three of the English shorts are about global warming.

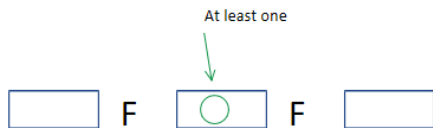
We are asked to arrange the play sequence so that:

The first and last movies are about global warming, and the two French movies are not placed back-to-back (there is at least one English movie between the French ones).

How many arrangements meet the requirements?

Start and end: $P(3,2) = 6$ Remove the two global warming movies from total.

Remaining: 8 movies (2 French, 6 English). Treat the 6 English movies as marbles.



$$m = 6 - 1 = 5 \quad n = 3 \quad C(5 + 3 - 1, 5) = C(7, 5) = 21$$

Final Answer: $P(3,2) \times 2! \times 6! \times 21 = 181440$

Note that $2!$ Permutes the two French movies, $6!$ Permutes the English movies, 21 takes care of the arrangement of the French movies in the sequence based on the requirements stated in the problem, and $P(3,2)$ is the number of ways we can start and finish the sequence.

13. How many ways are there to choose 10 co-op candidates out of 22 students to send to work at the same company if four of the students insist on being together (you can select all four of them or none of them) and another two students won't work together (you can select none of them or either one of them, but not both of them)? Hint: make a case-by-case argument, for example, one case is: employ all 4 friends and one of the enemies and 5 regulars ...

Case1: we hire all four friends, none of the two enemies, and 5 regulars:

$$C(4,4) \times C(2,0) \times C(16,6) = 8008$$

Case2: we hire all four friends, one of the two enemies, and 4 regulars:

$$C(4,4) \times C(2,1) \times C(16,5) = 8736$$

Case3: we hire none of the four friends, none of the two enemies, and 10 regulars:

$$C(4,0) \times C(2,0) \times C(16,10) = 8008$$

Case4: we hire none of the four friends, one of the two enemies, and 9 regulars:

$$C(4,0) \times C(2,1) \times C(16,9) = 22880$$

Add these disjoint cases up: 47632

14. Consider the equation $x_1 + x_2 + x_3 + x_4 = 17$

- A. How many positive integer solutions are there if the only restriction is that $x_i \geq 0$ Give **two different examples** of integer solutions for this equation.

Examples of solutions: $x_1 = 0, x_2 = 12, x_3 = 3, x_4 = 2$ or $x_1 = 0, x_2 = 1, x_3 = 0, x_4 = 16$

Number of all integer solutions: we have 17 marbles to distribute among 4 containers.

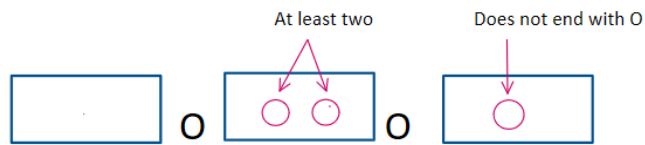
$$C(17 + 4 - 1, 17) = C(20, 17) = 1140$$

- B. How many integer solutions of the previous equation exist if we require each $x_i \geq 2$?

Put two marbles in each container and distribute the rest.

$$m = 17 - 2 \times 4 = 9, \quad n = 4 \quad C(9 + 4 - 1, 9) = C(12, 9) = 220$$

15. How many arrangements of the word **MICROBLOGGING** have at least three letters between the O's and do not end with O?



Total number of letters: 13

Marbles left: *all* – 2 *O's* – 2 *between O's* – 1 *in the end* = $13 - 5 = 8$

Containers open for business: $n = 3$

$$C(8 + 3 - 1, 8) = C(10, 8) = 45$$

For each of these 45 configurations, there will be $\frac{11!}{2!3!}$ permutations.

Final answer: $45 \times \frac{11!}{2!3!} = 149688000$

16. What percent of different arrangements of all the letters of the word ELEMENT have all the E's together? Example: TEEENML

All permutations: $\frac{7!}{3!} = 840$

The ones with E's together: tie up the E's together and treat as a new symbol, say θ

Number of permutations of $\theta, l, M, N, T = 5!$

The percentage of arrangements in question: $\frac{5!}{840} = 14.3\%$

17. BCIT offers two coding bootcamps for the students who do not have coding experience. One is a Java bootcamp and the other is Python. Each bootcamp is booked in a room that has a max capacity of 6 people. Eight students are randomly assigned to the bootcamps in such a way that all students have a place to go, and no room will be over capacity. In how many different ways can people be assigned to these two bootcamps?

Note that:

The bootcamps run at the same time and we cannot enroll the same student in both.

Java ABCD Python EFGH is a different case from Python ABCD JAVA EFGH

Java ABCD Python EFGH is the same case as Java CBDA Python GHFE

Solution by cases:

Java participants	Python participants	Number of choices
2	6	$C(8,2) = 28$
3	5	$C(8,3) = 56$
4	4	$C(8,4) = 70$
5	3	$C(8,5) = 56$
6	2	$C(8,6) = 28$
		Add all: 238

18. A certain financial institution issues credit cards that are numbered as follows: a 6-digit IIN (issuer identification number) followed by a 16-digit PAN (primary account number).

IIN has to be either between 222100 to 272099 (inclusive) or, between 510000 to 559999 (inclusive).

The PAN cannot be all zeros.

- a) How many valid credit card numbers can be generated following the above requirements?
- b) How many valid credit card numbers in part a) have a PAN that start and end with 7 and do not contain the digit 5 (no extra restrictions on IIN other than what is given in the problem).

a) IIN part:

Between 222100 to 272099 (inclusive), there are $272099 - 222100 + 1 = 50000$ possibilities, all of which have 6 digits. Also, between 510000 to 559999 (inclusive) there are $559999 - 510000 + 1 = 50000$ cases, all of which have 6 digits.

PAN part: $10^{16} - 1$ (all zeros is not allowed).

Total number of credit card numbers: $2 \times 50000 \times (10^{16} - 1) \approx 10^{21}$

b) IIN: same as before.

PAN: starts with 7 and ends with 7, so we need to count possibilities for the middle 14 digits.

Since 5 is no longer possible, each place can be filled in 9 ways: 9^{14}

The answer: $2 \times 50000 \times 9^{14} \approx 2.288 \times 10^{18}$

19. John Snow wants to give 13 Game Of Thrones shirts to six friends. The shirts are identical. In how many ways can he do that if:

a) There are no restrictions.

b) John's favorite friend (Samwel Tarley) is to receive at least 5 shirts.

a) Distribute 13 marbles into 6 containers: $C(13 + 6 - 1, 13) = C(18, 13) = 8568$

b) Give Samwel 5 shirts too begin with, distribute the remaining: $C(8 + 6 - 1, 8) = 1287$

20. As the manager in charge of making the weekly shift schedule, you are assigning four employees to the following grid:

Let's call the four employees A, B, C, and D (short for Ali, Brad, Carmen, and David)

Each employee gets exactly four shifts in total (could be on the same day or different days). Here is an example of what the schedule could look like:

	Shift 1	Shift 2	Shift 3	Shift 4
Monday	B	B	D	A
Tuesday	A	D	A	D
Wednesday	D	B	C	C
Thursday	B	C	A	C

a) In how many ways can you assign the shifts?

We can turn the above table into a string: BBDAADADBCCBCAC.

Number of ways to assign shifts is the same as the number of permutations of the above string:

$$\frac{16!}{4!4!4!4!} = 6306300 \text{ ways}$$

b) Ali has asked you to give him the first shifts on Monday and Tuesday. Other than this, he is indifferent towards any other slots, so he is okay to get any of the remaining shifts as long as he has a total of 4 shifts as stated above. Everyone else needs 4 shifts and have made no special requests. In how many ways can you set the schedule?

The strings will be of the form: A --- A -----

We have 14 empty slots that can be filled with 2A's, 4B's, 4C's, and 4D's:

$$\frac{14!}{2!4!4!4!} = 3153150 \text{ ways}$$

21. A computer system considers a string of decimal digits to be a valid code word if and only if the string is 6 digits long, it starts with an even number, and does not end in an even number. Repetition is allowed.

For example, **012227** is an acceptable sequence while **786588** is not an acceptable sequence.

a) How many 6-digit strings are acceptable?

$$\text{Even} \text{ --- Odd: } 5 \times 10^4 \times 5 = 250000$$

b) In what percent of the acceptable strings, there are exactly four even digits?

The first place is E (even)

The last place is O (odd)

We need three more places (among the middle 4) for even digits and we need to fill them with even digits; for this, we have $C(4,3) = 4$ choices.

Now consider one of those configurations:

EEEEEO

To fill up with actual digits, we have $5 \times 5 \times 5 \times 5 \times 5 = 5^5$ choices.

The answer is: $\frac{4 \times 5^5}{250000} \times 100\% = 25\%$

22. Each of the commands in line 6 and line 9 take 2 units of time to execute (each time). The *if-command* in line 5 takes 7 units of time to execute (each time). All other lines take an insignificant amount of time. How much time is needed for the entire code block to execute?

```
1 Counter = 7
2 Counter = 12
3 for (i = 1 to 400) {
4     for(j = i to 400) {
5         if( i≠j){
6             counter=counter+6
7         }
8         for (k=12 to 153){
9             counter=counter+12
10        }
11    }
12 }
```

Number of times line 6 is executed:

Note that for the line 6 to be executed, we need two distinct values, one for i, one for j: $C(400,2) = 79800$ times

Number of times line 5 is executed:

The logical condition in line 5 will be checked as many times as there are combinations of i and j. Note that this is selection with repetition:

$$m = 2, n = 400, \quad C(2 + 400 - 1, 2) = C(401, 2) = 80200$$

Number of times line 9 is executed:

For every (i, j) combination, we have a fixed number of k-instances: $153 - 12 + 1 = 142$, therefore the rule of multiplication applies: $80200 \times 142 = 11388400$

Total amount of time for the entire code block:

$$2 \times 79800 + 2 \times 11388400 + 7 \times 80200 = 23497800$$

23. Your friend has 10 coins in her pocket: Toonies, Loonies, Quarters and Dimes. If you correctly guess how many coins she has, you will win the coins.

- a. What is the probability that you guess the number of each coin type correctly?

Number of all possible guesses: $m = 10, n = 4, C(10 + 4 - 1, 10) = 286$

Number of correct guesses: 1

Probability of correct guess: $\frac{1}{286} \approx 0.35\%$

- b. As a hint, she tells you that she has at least four of her coins are quarters and asks you to guess the number of other coins. Find the probability of correctly guessing this time.

Put 4 marbles aside (they represent the quarters). Solve with the remaining marbles:

$$m = 10 - 4 = 6, \quad n = 4 \quad C(6 + 4 - 1, 6) = 84$$

Probability of the correct guess: $\frac{1}{84} \approx 1.19\%$